even in the distant future. But for kicks, here's how Zeilinger fancifully dreams we might one day go about it.

Imagine I want to teleport my DeLorean from New York to London. Instead of providing Nicholas and me with one member each of an entangled pair of photons (what we needed to teleport a single photon), we must each have a chamber of particles containing enough protons, neutrons, electrons, and so on to build a DeLorean, with all the particles in my chamber being quantum entangled with all those in Nicholas's chamber (see Figure 15.1). I also need a device that measures joint properties of all the particles making up my DeLorean with those particles flitting to and fro within my chamber (the analog of measuring joint features of Photons A and B). Through the entanglement of the particles in the two chambers, the impact of the joint measurements I carry out in New York will be imprinted on Nicholas's chamber of particles in London (the analog of Photon C's state reflecting the joint measurement of A and B). If I call Nicholas and communicate the results of my measurements (it'll be an expensive call, as I'll be giving Nicholas some 10^{30} results), the data will instruct him on how to manipulate the particles in his chamber (much as my earlier phone call instructed him on how to manipulate Photon C). When he finishes, each particle in his chamber will be in precisely the same quantum state as each particle in the DeLorean (before it was subjected to any measurements) and so, as in our earlier discussion, Nicholas will now have the DeLorean.[41] Its teleportation from New York to London will be complete.

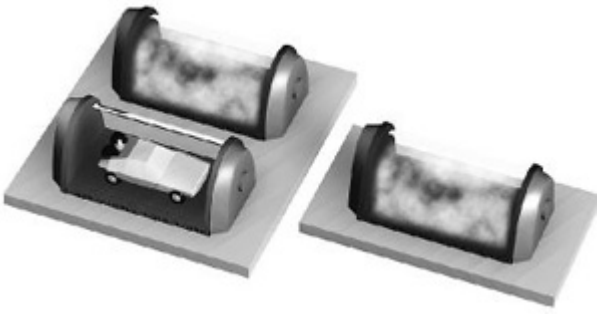


Figure 15.1 A fanciful approach to teleportation envisions having two chambers of quantum entangled particles at distant locations, and a means of carrying out appropriate joint measurements of the particles making up the object to be teleported with the particles in one of the chambers. The results of these measurements would then provide the